



Airplane Scatter Plugin

Quick Reference · FM-DX-Webserver · Author: Highpoint

v1.1 · April 2026

✈️ WHAT IS AIRPLANE SCATTER?

An aircraft at **8 000–12 000 m cruise altitude** reflects FM signals between a distant transmitter and your receiver — even when both are beyond each other's radio horizon.

The plugin **predicts these events in real time**, up to 3 min ahead, and displays them on an interactive map with a live countdown.

Best path lengths: **400–1 500 km** · Typical event duration: **15–60 s**

⚙️ HOW IT WORKS

- Downloads FM-TX database from **fmdx.org** (≤ 750 km, cached 24 h)
- Fetches live ADS-B positions every **15 seconds**
adsb.one → adsb.lol → adsb.fi (auto-failover)
- Evaluates scatter geometry for every **aircraft × TX pair**
- Applies radio-horizon, ellipse & elevation-angle gates
- Ranks results by **composite score 0–100 %**
- Shows top candidates with **ETA countdown** on map

📊 SCATTER SCORE (0–100 %)

Factor	Weight
Cross-track distance ($\sigma = 25$ km)	30 pts
Reflection angle (optimum 40° , $\sigma = 25^\circ$)	25 pts
Aircraft altitude (1 500–12 000 m)	15 pts
Fuselage alignment ($\perp$ bisector)	10 pts
TX ERP (logarithmic)	10 pts
FM frequency (87.5 vs 108 MHz)	5 pts

× Size multiplier: A5 heavy × **1.30** · A4 large × 1.15 · A3 × 1.00 · A2 × 0.85 · A1 × 0.70

Red ≥ 80 % Excellent **Orange ≥ 60 %** High
Yellow ≥ 40 % Medium **Green < 40 %** Low

🕒 ETA COUNTDOWN

Candidates appear **3 min before** and remain **1 min after** the crossing point.

–2:45 Approaching (blue)

NOW ✓ At crossing ± 5 s (green)

+0:38 Past crossing (red)

Dead reckoning fills the gap between 15-second ADS-B updates — icon & ETA refresh *every second* smoothly.

Score forecast evaluates **11 time steps** (0 – 300 s ahead); best score is used.

📡 DATA SOURCES

- **TX DB:** maps.fmdx.org
freq, ERP, city, country
- **ADS-B:** adsb.one / .lol / .fi
- **Elevation:** opentopodata.org
SRTM 90 m, batch, cached
- **Flags:** flagcdn.com
(ITU→ISO)
- **Rotator/GPS:** FM-DX-Webserver WebSocket

📏 GEOMETRY GATES

- TX–RX ≥ 400 km
- Elevation angle to AC $< 9^\circ$ (TX & RX)
- Both ends inside aircraft's **radio horizon**
- $d_{tx} + d_{rx} \leq 1.02\text{--}1.08 \times d_{direct}$
- Along-track $\leq d_{txrx} \times 1.1$
- Aircraft > 50 kts & $> 1\,000$ ft

🗺️ MAP PANEL

- ● Receiver QTH dot
- ● ● ● TX dots (sized by ERP)
- ✈️ Aircraft icon (rotated, colour = score)
- - - - Scatter path TX ↔ AC
💠 RX
- Left: candidate list sorted by |ETA|
- Bottom: terrain profile (interactive)
- Status bar: AC count · candidates · DB size

NEW IN V1.1

- **Multi-Select Compass:** click several directions at once
- **Rotor Sync Lock**  : compass auto-tracks PST Rotator — smart 22.5° boundary logic lights both adjacent sectors
- **Click-to-Turn Azimuth:** click bearing in TX detail → rotor turns (Admin/Tune required)
- **Frequency Sync Lock**  : map filters to receiver's tuned frequency
- **CPU Yielding:** large TX loads without audio stutter (RPi-safe)
- **Bounding-Box Pre-Filter:** faster fmdx.org JSON parsing
- Smart "Filtered by 1 station" clear (preserves other filters)

TIPS FOR BEST RESULTS

- Use **AI Denoiser** for weak scatter audio
github.com/Highpoint2000/AI-Denoise
- Use **RDS AI Decoder** for fragmented RDS bursts
github.com/Highpoint2000/RDS-AI-Decoder
- Heavy jets (A380, B747) = **+30 % score bonus** — prioritise!
- Low-power hardware: reduce Fetch Radius to 500 km, raise Min ERP to 100 kW
- Click  **Reload** to force a fresh TX database download
- Default threshold 75 % keeps noise low; lower it to see more candidates

KEY USER SETTINGS (VIA BUTTON IN PANEL HEADER)

Setting	Default	What it does
Min TX–RX Distance	400 km	Excludes nearby TX. Scatter needs both ends below radio horizon.
Min TX ERP	100 kW	Filters weak stations. Lower = more candidates, more CPU.
Lead / Trail Time	3:00 / 1:00	Window around crossing during which candidate is displayed.
Fetch Radius	750 km	Radius for TX database & ADS-B aircraft request.
Min Scatter Score	75 %	Minimum peak score required to create a candidate entry.
RX Antenna Height (AGL)	10 m	Your antenna height above ground — affects horizon & profile.
Unit System	Metric	Switch between km/m and miles/ft/knots.